

Microsoft Foundry AI Platform Workshop — Agenda

Format: Full-Day Led Demo (presenter-driven, audience observes)

Duration: ~7 hours

Scenario: Contoso Estimator Advisor — an AI agent for a fictional construction & engineering company that helps estimators prepare project bids.

Overview

A comprehensive enablement workshop covering the **Microsoft Foundry** platform — the unified Azure platform for enterprise AI operations, model building, and agent development. We progressively build a single production-grade agent across the day, adding capabilities module by module.

Scenario: Contoso Estimation Advisor

Throughout this workshop, we progressively build a single agent — the **Contoso Estimator Advisor** — for **Contoso Infrastructure**, a fictional large-scale construction and engineering company. The agent helps estimators prepare project bids by:

- Searching **historical project data** (past bids, final costs, lessons learned)
- Looking up **rate libraries** (labor, plant, materials by region)
- Referencing **company policies** (margin guidelines, approval thresholds)
- Performing **cost calculations** (preliminary estimates from BOQ quantities)

Agenda

Time	Module	Topic	What You'll See
09:00 – 09:45	Module 1	Foundry Platform Overview & Setup	Architecture, resource provisioning, GPT 5.4 deployment, RBAC
09:45 – 10:45	Module 2	Build Your First Agent (Low-Code)	Create the Contoso Estimator agent with File Search + Code Interpreter
10:45 – 11:00	Break		
11:00 – 11:45	Module 3	Agentic RAG & Foundry IQ	Multi-source knowledge bases, semantic retrieval, citation
11:45 – 12:30	Module 4	Content Safety & Guardrails	Prompt Shields, custom filters, block confidential data leakage
12:30 – 13:15	Lunch		
13:15 – 14:00	Module 6	Observability & Tracing	Application Insights, traces, Agent Monitoring Dashboard

Time	Module	Topic	What You'll See
14:00 – 14:45	Module 11	Pro-Code Development (.NET)	.NET SDK walkthrough — programmatic equivalent of the portal steps
14:45 – 15:00	<i>Break</i>		
15:00 – 15:40	Module 7	Evaluation & Continuous Monitoring	Batch eval, LLM-as-judge, CI/CD integration
15:40 – 16:15	Module 8	AI Gateway & Token Governance	APIM integration, TPM limits, cost control, semantic caching
16:15 – 16:30	Wrap-up	Q&A and Next Steps	Discussion, customization options, follow-up resources

Key Outcomes

By the end of the day, attendees will understand how to:

1. Provision and configure a Microsoft Foundry resource
2. Build, ground, and secure production-grade agents
3. Integrate enterprise knowledge with Foundry IQ
4. Apply content safety and Responsible AI guardrails
5. Pro-code version of building the agent
6. Monitor agents with tracing and continuous evaluation
7. Govern AI usage with AI Gateway (rate limits, cost control)

Format & Prerequisites

- **Led Demo** — presenter demonstrates each module live; no setup required for attendees
- Bring questions specific to your organization's AI use cases
- Optional pre-reading: [What is Microsoft Foundry?](#)

Optional Add-On Modules

Available on request to tailor the agenda:

Module	Duration	Topic
Module 5	45 min	Foundry Toolkit & Data Connectors (MCP, Fabric, SharePoint, OpenAPI)
Module 9	50 min	Safety & Red Teaming (automated adversarial testing)
Module 10	50 min	Fleet Management & Agent 365 (Control Plane, publish to Microsoft Teams)